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WaterMAP: A Scalable Machine Learning Framework for Emission-Factor-Derived Spatiotemporal GHG Prediction and Mitigation in Wastewater Treatment Plants

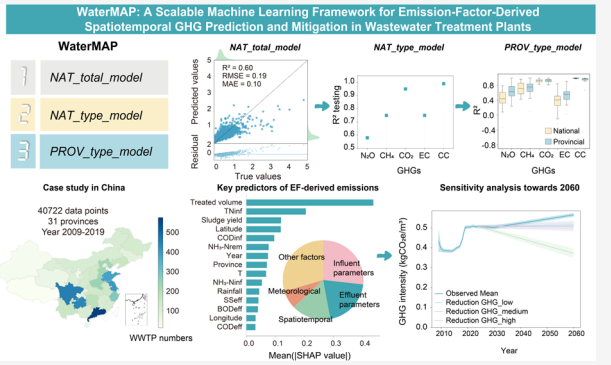
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ABSTRACT: Greenhouse gas (GHG) emissions from wastewater treatment have gained increasing attention in global climate governance. However, conventional emission-factor (EF)-derived inventories lacked the ability to capture nonlinear variations, while existing machine learning (ML) models remained fragmented in scale. Here, we introduce WaterMAP (**W**astewater **AI** **T**reatment **E**mission **R**egression for **M**ulti-scale **A**ccounting and **P**rediction), a novel ML framework designed for spatiotemporal EF-derived GHG prediction and mitigation evaluation. Using 40,722 inventory-based GHG records from 5155 WWTPs in China during 2009−2019 as a case study, the *National_Total_Model* achieved a test RMSE of 0.19 kg CO_2-e/m^3 when predicting the total EF-derived intensity. *National_Type_Model* showed that scope 1 totaled 7.6 Mt CO_2-e, scope 2 accounted for 18.9 Mt CO_2-e, and scope 3 contributed 0.5 Mt CO_2-e in 2019. We then estimated cumulative GHGs for 2024 to be 36.8 Mt CO_2-e, with an average intensity of 0.51 kg CO_2-e/m^3. Treated volume, TN_{inf}, sludge yield, latitude, and COD_{inf} were identified as the key predictors influencing EF-derived GHG emissions. We further proposed a GHG spatial heterogeneity index using *Provincial_Type_Model*, reflecting urban development. Under exploratory sensitivity analysis toward 2060, WaterMAP-guided projections suggested 9.6−34.0% potential reduction. Overall, WaterMAP offers a scalable framework for EF-derived GHG emissions screening and prediction, supporting GHG mitigation assessment.



KEYWORDS: *greenhouse gas emission, wastewater treatment, machine learning, EF-derived inventory, prediction and mitigation*

INTRODUCTION

Achieving carbon neutrality and advancing the development of zero-carbon cities have emerged as central objectives in global climate governance.[1] According to the Intergovernmental Panel on Climate Change (IPCC), the role of wastewater sector-related greenhouse gases (GHGs) has increasingly received attention, with annual global emissions of 0.38 Gt CO_2-e, comparable to those of the chemical industry (0.37 Gt CO_2-e).[2] In the context of zero-carbon cities, wastewater treatment plants (WWTPs) serve a dual role as both GHG sources and resource recovery nodes. Accurate GHG accounting across scopes 1−3 is, therefore, the first and essential step for guiding highly efficient treatment and development of water resource recovery facilities (WRRFs). Scope 1 (direct) emissions from WWTPs mainly involve nitrous oxide (N_2O), methane (CH_4), and fossil carbon dioxide (CO_2),[3] with CH_4 and N_2O ranking as the fifth and third largest anthropogenic sources among the major GHG-contributing industries, respectively.[4] Additionally, the energy consumption (scope 2, indirect, EC) is substantial, accounting for an estimated 0.8−4% of global electricity usage.[5] Accounting and reporting for scopes 1−2 emissions are mandatory, whereas scope 3 (including construction, chemicals, and byproduct use, indirect) reporting remains optional.[4] The incomplete coverage underscores the urgent need for robust and comprehensive carbon accounting frameworks capable of capturing the full life-cycle footprint of wastewater systems and thereby informing effective emission reduction pathways.[6]

Current accounting methods include direct measurement, emission factor (EF) method, and modeling strategies.[7−9] Direct measurement offers high accuracy but is constrained by high costs, intensive labor requirements, and the complexity of

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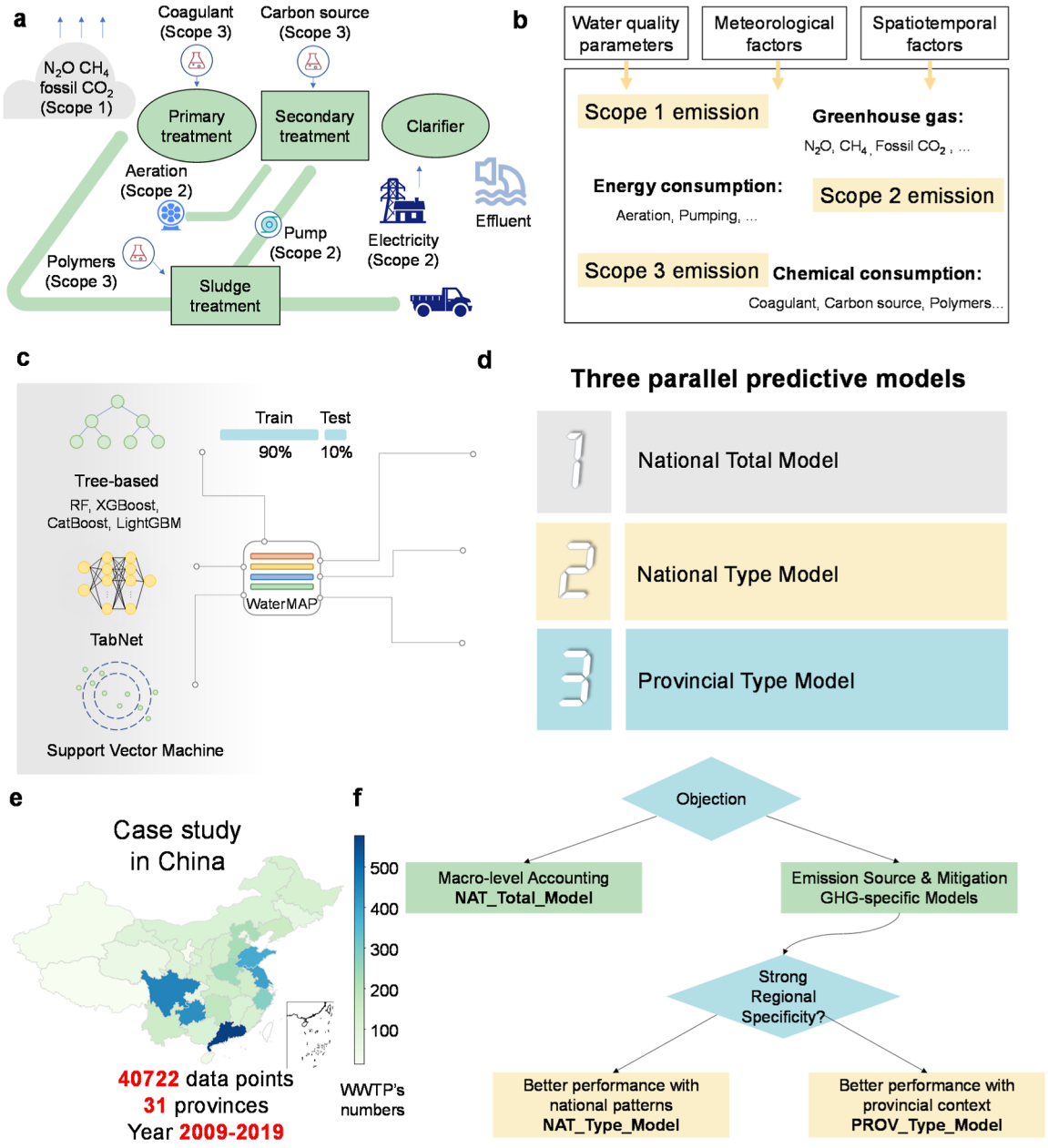


Figure 1. The WaterMAP framework for predicting GHGs emissions from WWTPs. (a) The pathway of GHG (scope 1−3) emissions from WWTPs. (b) The inputs and outputs of machine learning models. (c) The six ML models including tree-based (RF, XGBoost, CatBoost, and LightGBM), TabNet, and SVM. (d) Three parallel predictive models. (e) Case study in China and overview of data points. (f) Model selection for different conditions.

covering large-scale WWTPs.[10] IPCC guidelines focused on the EF method combined with activity data, which enabled large-scale accounting and was successfully applied in diverse countries and regions (such as China and the US) due to the simplicity and reliability.[3,6] Although the EF method facilitates GHG emission estimation and accounting, it cannot effectively identify environmental impacts on emissions or characterize their inherent nonlinear dynamics. On the other hand, modeling approaches have been categorized into mechanistic and data-driven models. Mechanistic models (such as ASM, activated sludge model) relied on Monod equations and several implicit parameters, which lacked the flexibility when changing environmental conditions.[11] Besides, carbon emissions were affected by high-dimensional environmental factors (such as influent C/N ratio, temperature, and so on) and spatiotemporal disparities in socioeconomic development, meaning that any system shifts required recalibration of mechanistic model parameters.

In comparison, machine learning (ML) algorithms could develop wastewater management models based solely on inputs and outputs even under high-dimensional environmental conditions.[12−14] Huang et al. used data-driven models to predict facility-scale energy-related carbon emissions in China's large-scale WWTPs, achieving better results than mechanism-based mass balance models.[15] Despite these advances, ML applications in the wastewater sector still remained constrained by several limitations. First, existing studies focused on the single GHG type/spatial scale/plant, which might limit the ability to simultaneously support multigranularity decision-making including national totals for

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Table 1. WaterMAP Framework for Predicting EF-Derived GHG Inventories from WWTPs

models	*NAT_total_model*	*NAT_type_model*	*PROV_type_model*
purposes	total GHG prediction	specific GHG prediction and identification of key predictors	province-level GHG prediction and transferability; calculation of GHG Spatial heterogeneity index (GHG_SHI)
input variables	(1) influent parameters: chemical oxygen demand (COD_{inf}), biochemical oxygen demand (BOD_{inf}), total nitrogen (TN_{inf}), ammonia nitrogen (NH_3–N_{inf}), total phosphorus (TP_{inf}), suspended solids (SS_{inf}), COD/TN ratio (C/N), COD/TP ratio (C/P), BOD/COD ratio (B/C) (2) effluent parameters: effluent COD, BOD, SS, NH_3–N, TN, TP (COD_{eff}, BOD_{eff}, SS_{eff}, NH_3–N_{eff}, TN_{eff}, TP_{eff}), COD/BOD/SS/NH_3–N/TN/TP. Removal (COD_{rem}, BOD_{rem}, SS_{rem}, NH_3–N_{rem}, TN_{rem}, TP_{rem}) (3) spatiotemporal factors: year, longitude, latitude, province (4) meteorological factors: annual provincial mean rainfall, temperature, relative humidity (5) other factors: treated volume, sludge yield		same as *NAT_total_model* excluding province, longitude, and latitude
labels/outputs	total EF-derived volumetric GHG intensity	volumetric EF-derived GHG intensity of five types (N_2O/CH_4/fossil CO_2/EC/CC)	same as *NAT_type_model* but trained separately for each province
users	national accounting and policy support	typology-based identification of reduction pathways	provincial operation and interprovincial transfer analysis

climate commitments, GHG-specific decomposition for operational control, and regional resolutions for action plans. While emissions could be aggregated or disaggregated across scales, such transitions might introduce additional uncertainties or systematic biases. Second, ML and EF approaches have often been presented as alternative pathways, whereas we consider them potentially complementary for large-scale applications where observational data remain limited. EF methods could provide retrospective accounting based on activity data, while ML enabled cross-scenario EF-derived emission prediction and quantified the influence of key predictors. Third, another gap lay in leveraging WWTP GHG emissions as dynamic feedback signals for urban development to reflect broader urban metabolic states under varying climatic, socioeconomic, and technological conditions.[16] We, therefore, aimed to address these gaps by integrating comprehensive scope 1–3 emissions and deciphering key factors—GHG inventory—urban development coupling.

This work introduced a methodological ML framework, WaterMAP (**W**astewater **AI** **T**reatment **E**mission **R**egression for **M**ulti-scale **A**ccounting and **P**rediction) (Figure 1), which integrated the EF method and data-driven models. WaterMAP comprised a family of parallel predictive models: *National_Total_Model* (*NAT_Total_Model*) for aggregate national accounting, *National_Type_Model* (*NAT_Type_Model*) for GHG type-resolved attribution, and *Provincial_Type_Model* (*PROV_Type_Model*) for province-resolved regional analysis, ensuring that predictions at each granularity were directly optimized, avoiding errors from aggregation or disaggregation. Explanatory techniques were utilized to interpret the complex and nonlinear relationships between key predictors and GHG inventory across multiple scales.[14] Subsequently, we conceptualized GHG emissions of WWTPs as an urban carbon sink and established correlations between emission patterns and urban development. Under China's dual-carbon policy, where stringent effluent standards and low-carbon treatment requirements were already established and geographical factors significantly influenced carbon emissions, we selected China as the case study to validate the proposed framework and conduct decarbonization analysis.

MATERIALS AND METHODS

The overall architecture of WaterMAP and modeling pipeline is illustrated in Figure 1.

Variables and Data

Water quality, quantity, and sludge yield data for 2009–2019 were obtained from the Ministry of Housing and Urban-Rural Development of China (MOHURD),[3] which represented pollutant loads driving GHG formation and captured treatment performance. Process data were only recorded for 2019, and ML techniques were applied to link water quality and process to reconstruct data for the prior ten years (Text S8). Meteorological variables (rainfall, temperature, and relative humidity) were incorporated for their effects on bioreaction kinetics and dilution from China's National Bureau of Statistics. Spatiotemporal factors were used in *NAT_total_model* and *NAT_type_model* to capture regional and temporal heterogeneity, whereas *PROV_type_model* only included year, reflecting temporal evolution within provinces (Table 1).

Because of the limited availability of measured GHG data at national scales, the EF approach was adopted as a surrogate for label generation. EF data were also sourced from the IPCC and published literature.[17] COD removal was used as the activity data for CH_4 and CO_2 emissions, with CO_2 further divided into fossil and biogenic components at a 10% ratio, while TN_{inf} served as the activity data for N_2O. EC was recorded by the MOHURD for scope 2 emissions. For scope 3, chemical consumption (CC) was considered as the dominant operational emission source and estimated from nutrient (N and P) removal (Text S1). GHG emissions were calculated using the EF method across different years, provinces, and facility-specific conditions to form the data set, including 36 variables and 40,722 data points (Table S54).

Model Development and Evaluation

NAT_total_model was designed to estimate EF-derived total volumetric GHG intensity, and *NAT_type_model* was developed to target distinct GHG types, while *PROV_type_model* was used to predict each GHG type at the provincial scale. The entire data set was split into a training set (90%) and a testing set (10%) using province-stratified sampling to avoid data leakage and ensure consistent data partitions across all three models (Figure S63). Province data were encoded using encoding techniques, and normalization was applied to the training set, with the same scaling factors used on the testing set (Texts S9 and S10). Six ML and DL algorithms (XGBoost, CatBoost, LightGBM, RF, SVM, and TabNet) were employed, leveraging their performances and interpretation in handling tabular data and environmental issues (Text S11).[18] Five-fold cross-validation and Bayesian optimization were used to tune hyperparameters, selecting the best-performing configuration based on cross-validation performance (Text S12). R^2 and RMSE were applied as evaluation metrics and evaluated on testing sets (Text S13). In addition, the learning curve (LC) was used to evaluate how model performance changed with increasing training sample size, thereby assessing the potential overfitting or underfitting.

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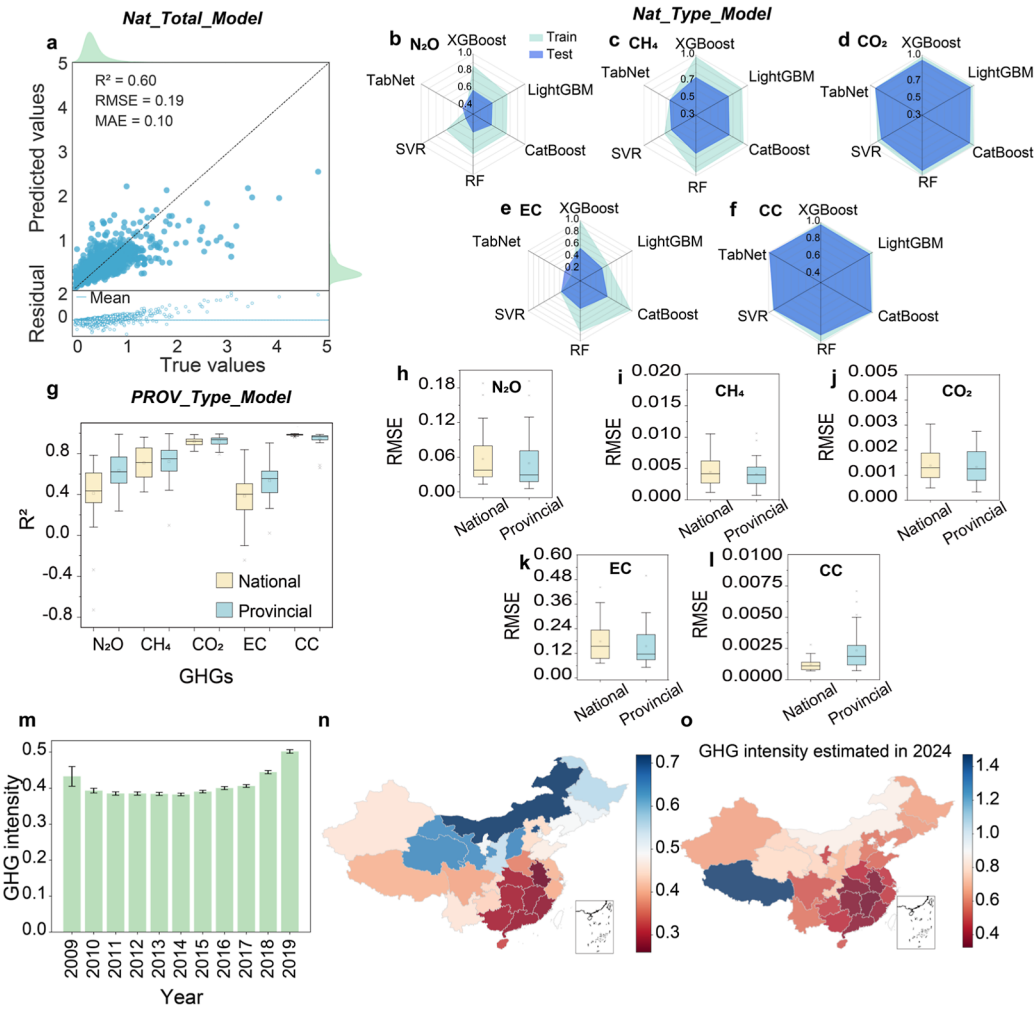


Figure 2. Results of EF-derived GHGs emission prediction and accounting. (a) The best-performing XGBoost model when predicting the total GHG emissions. (b–f) The training and testing R^2 of six ML models for each GHG type, including N_2O, CH_4, fossil CO_2, EC, and CC. (g–l) Boxplots of testing R^2/RMSE for different provinces regarding direct/indirect GHG emissions. (m,n) Accounted average GHG intensity across years and provinces. (o) GHG estimation in 2024.

Model Interpretation

SHapley additive exPlanations (SHAP) was applied to quantify the feature contributions on predicted GHG emission (Text S14), while partial dependence plots (PDP) characterized the factor sensitivity ranges between key predictors and GHG emissions.[18] Structural equation modeling (SEM), multicollinearity analysis, and partial correlation analysis were employed to assess the statistical associations underlying spatial heterogeneity (Texts S3 and S4). SEM included both direct and chain-mediated paths by standardized association coefficients using the *lavaan* package in R with predicted GHG intensity and key predictors as inputs. Unsupervised learning methods, including t-distributed stochastic neighbor embedding (t-SNE), were applied to investigate the influence of treatment processes on predicted GHG intensity.

GHG_SHI

Beyond conventional metrics focusing solely on accounting or model accuracy, testing RMSE from each province was selected to calculate GHG Spatial Heterogeneity Index (GHG_SHI).[16] Testing RMSE captured the portion of variability not explained by structured features, and normalizing it by the province-specific emission range reflected the contribution of unstructured spatial heterogeneity (e.g., local operational differences). For cross-province comparability, GHG_SHI was defined as the test RMSE divided by the interquartile range (IQR) of provincial emissions (eq 1). IQR was chosen to remove dependence on absolute emission levels and reduce sensitivity to extreme values. Lower GHG_SHI values indicated more homogeneous and predictable emissions explained by structured features, while higher values indicated significant unstructured spatial heterogeneity.

$$\mathrm{GHG_SHI}_p = \frac{\mathrm{RMSE}_p}{\mathrm{IQR}(y_p)} = \frac{\sqrt{\frac{1}{n_p}\sum_{i\in p}(y_i - \hat{y}_i)^2}}{Q_3(y_p) - Q_1(y_p)} \quad (1)$$

Where, n_p is the number of test samples in Province p, Q_1 and Q_3 are the 25th and 75th percentiles of GHG emissions, repsectively, and y_i and $\hat{y}_i$ are the true and predicted GHG values, respectively.

Regression analysis and Pearson correlation coefficient analysis were conducted between 16 urban indicators and GHG_SHI, which were categorized into four types: climate and environmental conditions, water resource utilization and management indicators, environmental sustainability and risk management parameters, and socioeconomic development and infrastructure investment indicators.

Scenario Analysis

(1) GHG estimation for 2020–2024. Provincial population data for 2020–2024 were integrated, with 2019 as the baseline, to estimate changes in treated volumes nationwide.[19] Incorporating 2009–2019 WWTP statistics,[20,21] Monte Carlo simulations under grade-1A standards were used to calculate GHG emission parameters and estimate emissions for 2020–2024.[22] (2) Model-guided exploratory

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sensitivity analysis toward 2060. Key predictor ranges were derived to construct low-, medium-, and high-intensity control scenarios, systematically analyzing long-term trends. (3) Cross-province transfer learning. For the best-performing *PROV_type_model*, one province was used as the source province for training using the data and optimal hyperparameters, subsequently tested on another target province. The procedure was repeated across all 31 provinces to generate a 31 × 31 cross-province transfer matrix, and GHG emission pattern transferability was quantified based on model performance (e.g., R^2 and RMSE) in the target provinces.

Uncertainty Analysis

For *NAT_total_model* and *NAT_type_model*, 1000 Monte Carlo simulations were performed given the inherent EF uncertainties (Text S5).[23] Scope 1 emissions were assigned differentiated and asymmetric uncertainty ranges for each process (Table S45), while scope 2−3 employed uniform ranges.[24] EFs were perturbed and labels updated to quantify the impact of parameter uncertainty on EF-derived emission estimates, model performance, and feature importance. For cross-province transfer of *PROV_type_model* and the uncertainty of the GHG_SHI, 1000 Bootstrap resamples were used to generate predictive or index distributions. The 2.5% and 97.5% percentiles were used to construct 95% confidence intervals (CI). CIs for *PROV_type_model* evaluated the stability and reliability of cross-province predictions, while CIs for GHG_SHI quantified the uncertainty of province-level spatial heterogeneity, with CI width representing the level of uncertainty.

Ablation Analysis

To evaluate the contribution of EF-related information to ML, we designed a systematic ablation framework across the *NAT_total_model* and *NAT_type_model*. EF-related activity data included direct activity data (COD_{rem}, TN_{inf}, TN_{rem}, TP_{rem}), extended process descriptors (e.g., influent/effluent concentrations), and derived stoichiometric ratios (C/N, C/P), and we constructed multiple model variants by selectively removing or retaining the EF variables for comparison (Text S15).

Generalization Analysis

(1) Leave-one-year-out (LOYO) validation: temporal generalization was examined by reserving 2019 as an unseen year for evaluation while fitting the best-performing models using data from 2009 to 2018 (Text S6).[25] (2) Leave-one-province-out (LOPO) validation: spatial generalization was evaluated through a 31-fold cross-validation procedure, where data from one province were excluded for evaluation and the models were fitted using data from the remaining provinces. (3) External sample validation: real operational data and EF-derived labels from 10 WWTPs in Hubei for 2024 (independent of the study data set) were used.[26] After the same data preprocessing, models were applied directly to predict emissions. Model performance in these unfamiliar scenarios was evaluated using the mean absolute percentage error (MAPE), and the adaptability differences between national and province-level models were compared.

RESULTS AND DISCUSSION

WaterMAP Framework Performance to Predict the EF-Derived GHG Emissions

NAT_Total_Model Performances. The training RMSE ranged from 0.03 to 0.35 kg CO_2-e/m^3 for XGBoost, CatBoost, LightGBM, RF, SVM, and TabNet, while the test RMSE was 0.19−0.25 kg CO_2-e/m^3 (TableS S1 and S2).[27] XGBoost performed the best (Figure 2a), achieving the highest test R^2 (0.60) and the lowest RMSE (0.19 kg CO_2-e/m^3). Uncertainty analysis indicated that the coefficient of variation (CV) was 8.90%, lower than EF uncertainty itself (e.g., A^2O N_2O: −97% to 456%). Retraining under EF label perturbations yielded R^2 = 0.64 ± 0.03, indicating that model performance remained relatively stable, supporting the robustness of EF-based inventory prediction at the national scale.[15] Linear regression yielded only R^2 = 0.12−0.19, further demonstrating the validity of EF-derived labels for ML. Moreover, removing activity data used in EF calculations resulted in only a negligible performance loss, with the model retaining 87.6% of its predictive power, which highlights the strong predictive capability of non-EF-formula features (Table S55). However, residuals revealed underestimation above 2 kg CO_2-e/m^3, likely due to the coexistence of multiple emission pathways, motivating the development of *NAT_type_model* for GHG type-resolved attribution.

NAT_Type_Model Performances. Figure 2b−f summarized the predictive performances of six ML models for five GHG emissions (Table S3). XGBoost excelled for predicting EC and N_2O ($RMSE_{test}$ = 1.12 and 1.3 kg CO_2-e/m^3, R^2_{test} = 0.58 and 0.54). LightGBM outperformed for CO_2 (0.23 kg CO_2-e/m^3, R^2_{test} = 0.95), CatBoost was most effective for CH_4 (0.44 kg CO_2-e/m^3, R^2_{test} = 0.75), and TabNet excelled in predicting CC (0.12 kg CO_2-e/m^3, R^2_{test} = 0.99), which partly showed better performances than *NAT_total_model*. Retraining under EF perturbations showed stable R^2 for EF-derived EC (0.53 ± 0.01), CO_2 (0.86 ± 0.08), and CH_4 (0.72 ± 0.08), with higher variance for N_2O (0.63 ± 0.07) reflecting its wider EF uncertainty. National-total CVs followed the order: EF-derived EC (5.7%) < CC (15.2%) < CO_2 (18.5%) < N_2O (39.4%) < CH_4 (42.4%), indicating different levels of sensitivity to EF perturbations across emission components. Figure S1 showed varying provincial predictions and emission patterns, with EC and N_2O performing the worst, even including negative R^2 in Hainan and Tibet. To improve the accuracy of province-resolved regional analysis, *PROV_type_model* was developed using the same methodology and splitting data sets.[20]

PROV_Type_Model Performances. All models underwent a new round of hyperparameter tuning to adapt to provincial data sets (Tables S4−S34), allowing the *PROV_type_model* to assess performance under provincial contexts. Tree-based models, particularly XGBoost and CatBoost, often achieved the best results (Figures S2−S6).[28] Overall, except for CC emissions, the best-performing *PROV_type_model* outperformed the *NAT_type_model* in terms of R^2 and RMSE (Figure 2g−l). For instance, provincial RMSE for N_2O decreased from 0.057 ± 0.044 to 0.049 ± 0.047 kg CO_2-e/m^3 and for EC from 0.178 ± 0.099 to 0.155 ± 0.094 kg CO_2-e/m^3. A few provinces showed slight R^2 declines (e.g., Sichuan for CO_2 and CC emissions), suggesting that the *NAT_type_model* has benefited from cross-province learning. Nonetheless, the *NAT_type_model* sometimes better captured unique provincial emission patterns, and the choice between *NAT/PROV_type_model* should depend on performance and province-level characteristics (Figure 1f). Besides, the learning curves showed that, with increasing training sample size, the test R^2 generally increased and/or the RMSE decreased across most models and provinces (Figures S7−S11), indicating improved generalization performance and no obvious overfitting.

WaterMAP-Assisted Emissions' Quantification

GHG Estimation for 2009−2019 Using WaterMAP. *NAT_total_model* provided robust national-level GHG estimates and a baseline for spatiotemporal comparison during 2009−2019 in China (Figure 2m). Temporally, the national volumetric GHG intensity followed a "decline-stabilization-

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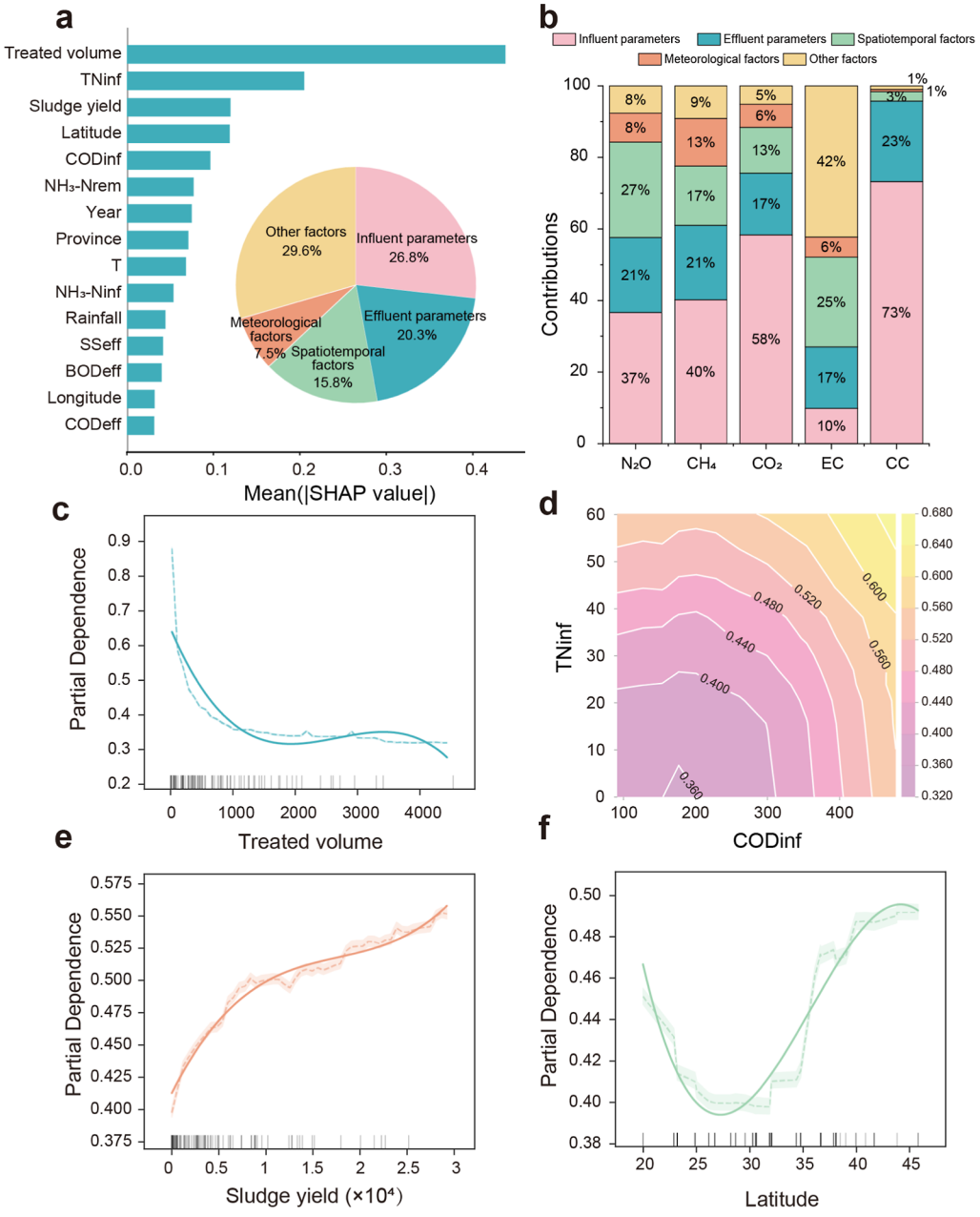


Figure 3. Results of national model prediction and interpretation. (a) The importance ranking of the best-performing XGBoost model when predicting the total GHG emissions. (b) The importance ranking (five categories) of six ML models on each EF-derived GHG, including N_2O, CH_4, fossil CO_2, EC, and CC. Partial dependence plots or 2D-PDP on (c) treated volume. (d) TN_{inf} and COD_{inf}. (e) Sludge yield. (f) Latitude.

rebound" pattern: decreasing from approximately 0.43 to 0.38 kg CO_2-e/m^3 (95% CI [0.38, 0.63] to [0.33, 0.58]) from 2009 to 2014, stabilizing through 2017, then rebounding to ~0.50 kg CO_2-e/m^3 (95% CI [0.44, 0.70]) in 2019.[23] Cumulative GHG increased from 9.28 Mt CO_2-e (95%CI [7.96, 14.66]) to 27.06 Mt CO_2-e (95%CI [23.56, 40.24]).[29] *NAT_type_model* quantified source-specific emissions (Figure S15): scope 1 totaled 7.6 Mt CO_2-e, scope 2 accounted for 18.9 Mt CO_2-e, and scope 3 contributed 0.5 Mt CO_2-e in 2019, highlighting EC as the dominant contributor to WWTP carbon emissions (Figure S16).[30]

Spatially, *NAT_total_model* revealed substantial regional heterogeneity in GHG estimation. Provinces with both high emissions and high uncertainty include Beijing (0.71 kg CO_2-e/m^3, CV = 15.51%), Inner Mongolia (0.72 kg CO_2-e/m^3, 14.57%), Shanxi (0.65 kg CO_2-e/m^3, 16.87%), and Shandong (0.47 kg CO_2-e/m^3, 17.30%) during 2009−2019 (Figures 2n and S26). During 2009−2019, *NAT_type_model* also showed Guangdong had the highest cumulative N_2O/EC emissions (4.0/13.9 Mt CO_2-e, 65.99%/7.73%) and Shandong had the highest cumulative CH_4/CO_2/CC emissions (0.9/0.5/0.4 Mt CO_2-e, 52.53%/28.88%/48.45%), highlighting regions of elevated emissions and uncertainty warranting focused attention (Figures S29−S34).

GHG Prediction in 2020−2024 Using WaterMAP. Cumulative GHG emissions rose from 30.1 Mt CO_2-e in 2020 to 37.6 Mt CO_2-e in 2023, followed by a slight decline to 36.8 Mt CO_2-e in 2024, with a volumetric GHG intensity of 0.51 kg CO_2-e/m^3 (Figures 2o and S14). While the intensity was comparable to that of the 2019 level, total emissions

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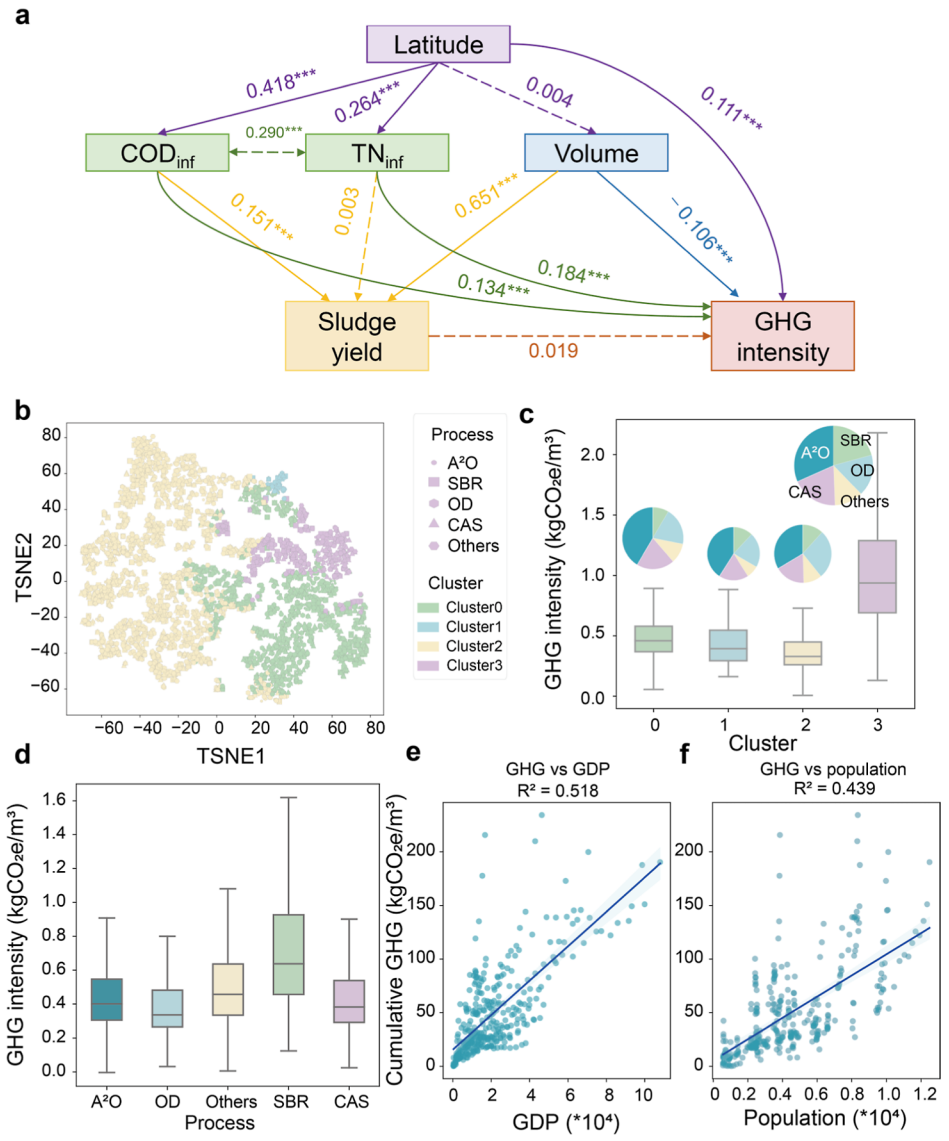


Figure 4. Results of latitude, process, population, and GDP influencing EF-derived GHG intensity. (a) The SEM of latitude influencing GHG intensity with the key predictors. (b) The t-SNE of different process on GHG intensity. (c) The boxplots of GHG intensity for four clusters and the process proportion. (d) The boxplots of GHG intensity for different processes. (e,f) The linear relationship of GDP/population with cumulative GHG intensity.

continued to show an upward trend. *NAT_type_model* prediction showed similar GHG type properties between 2019 and 2024, with scope 2 emissions accounting for the largest at 27.3 Mt CO_2-e. Meanwhile, the spatiotemporal disparities during 2020−2024 remained significant (Shandong and Guangdong exhibited the highest cumulative GHG emissions, Figure S17), with the Northwest region showing relatively higher volumetric intensity and the Central region showing comparatively lower volumetric intensity.[30] The spatiotemporal features highlighted the need to clarify the key predictors of carbon emissions and the necessity for *PROV_type_model* development.

Deciphering Key Predictors and Mitigation Pathways

Identifying Key Predictors Influencing GHG Intensity. Well-trained ML models enabled the prediction and accounting but failed to uncover the relationships between key predictors and EF-derived emissions.[31,32] The *NAT_total_model* identified macrolevel key predictors, while the *NAT_type_model* served as complementary for the specific GHG types.

Figure 3a illustrates the ranking of environmental factors influencing EF-derived inventories based on the *NAT_total_model*. Overall, influent water quality parameters, treatment performance (including effluent quality), and other operational factors (including treated volume) accounted for the major contribution (76.7%). Specifically, volume (23.24%), TN_{inf} (10.9%), sludge yield (6.35%), latitude (6.31%), and COD_{inf} (5.12%) were identified as the key predictors of EF-derived emissions of the total C emissions. In the uncertainty analysis, these five parameters consistently appear with frequencies between 76% and 100%, indicating model robustness under EF uncertainty (Figures S35 and 36). Figure 3b further shows the relative contributions of different factors to individual GHG types (*NAT_type_model*): EF-derived scope 1 and scope 3 emissions were mainly influenced by water quality parameters (58%−73%), while EF-derived scope 2 emissions were predominantly affected by treated volume (33%). PDPs

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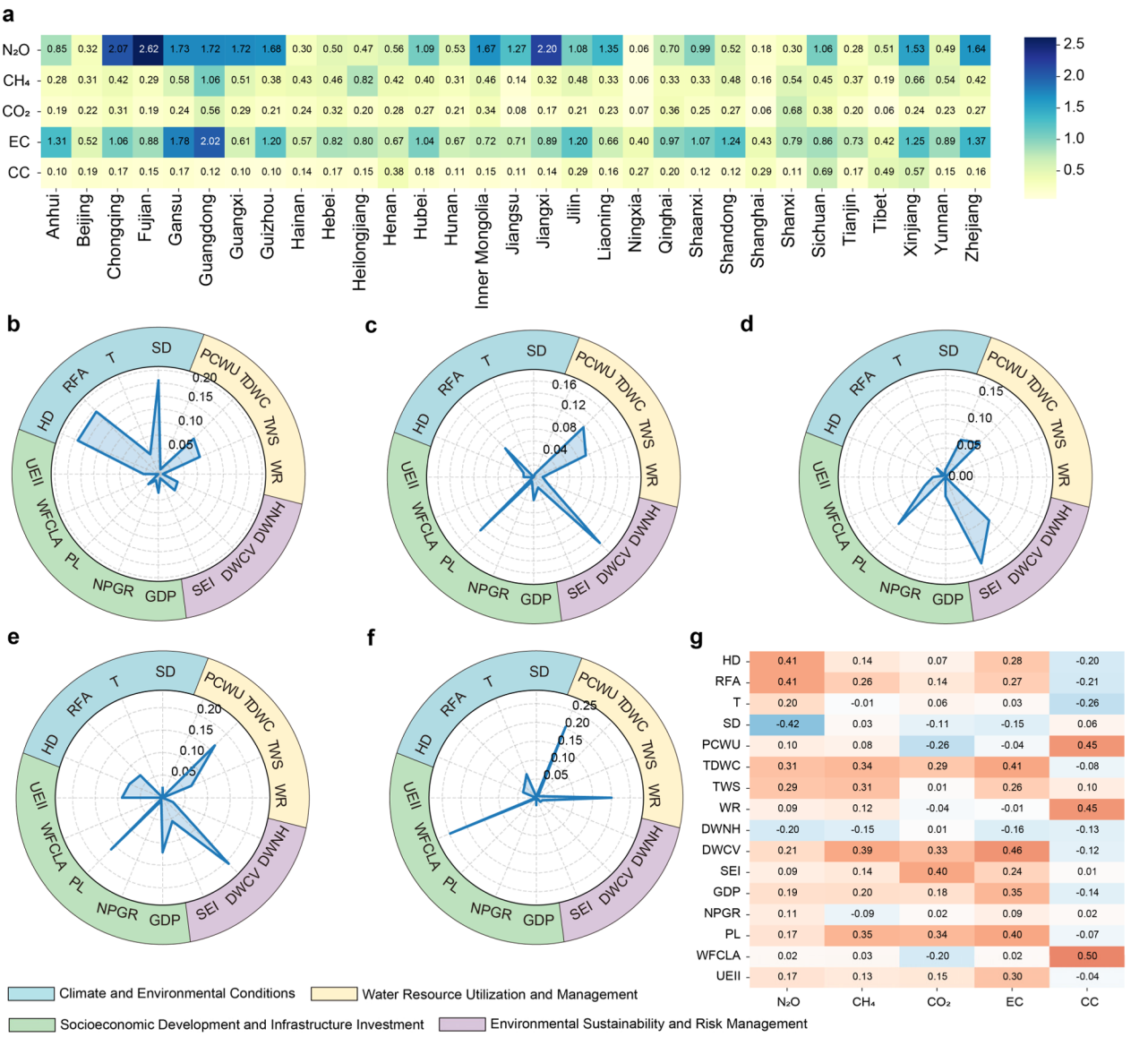


Figure 5. *PROV_type_model*-assisted GHG_SHI with urban development indicators: (a) the GHG_SHI across each province; socioeconomic development revealed by the intraprovince GHG emission spatial homogeneity (GHG_SHI). (b–f) Metrics (R^2 values) for the linear regression between 16 variables and the testing R^2 on different provinces of direct/indirect GHGs emission. (g) Metrics (Pearson correlation coefficient) between 16 variables and the GHG_PII on different provinces. HD, humidity; RFA, rainfall; T, temperature; SD, sunshine duration; PCWU, per capita water usage; TDWC, total domestic water consumption; TWS, total water supply; WR, water resources; DWNH, domestic waste nonhazardous treatment rate; DWCV, domestic waste collection volume; SEI, sudden environmental incidents; GDP, gross domestic product; NPGR, natural population growth rate; PL, population; WFCLA, water facility construction land area; and UEII, urban environmental infrastructure investment.

(Figure 3c–f) were utilized to further analyze the critical factors influencing EF-derived GHG inventories and identify specific ranges.

Insights from Load Buffering. The importance of treated volume, COD_{inf} and TN_{inf} underscored the significance of load management inspired by water quality and quantity dynamics. Figure 3c showed an increase in treated volume significantly reduced predicted GHG emissions, indicating a clear scale effect. As the treated volume expanded, the fixed energy costs were distributed across a larger water flow, thereby reducing GHG emissions. When the yearly treatment capacity exceeded 10–20 million m^3, the carbon emission curve declined markedly and then stabilized at higher volumes, consistent with Huang et al.[15] Similarly, the dominance of treated volume in EC emissions (33%) reflected correlations captured by the EC *NAT_type_model*.

Influent water quality parameters, especially the combined effects of COD_{inf} and TN_{inf}, were correlated with predicted emissions. The 2D-PDP plots (Figure 3d) showed that COD_{inf} of 100–300 mg/L and TN_{inf} of 0–25 mg/L were associated with lower predicted emissions (mean reduced from 0.83 to 0.36 kg CO_2-e/m^3). N_2O emissions had an almost linear increase with TN_{inf}, consistent with nitrification/denitrification kinetics, where higher nitrogen provides more substrate for microbial N_2O production[33,34] (Figure S19 and Text S2). Even though the EF labels were based on COD_{rem}, CH_4 and fossil CO_2 still shared similar dominant predictors (primarily COD_{inf}, ranked as the third and second), validating the role of organic carbon availability on methanogenesis and CO_2

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generation.[35] Although WWTPs could not easily control influent composition, both existing and future facilities could employ hydraulic and quality equalization (e.g., equalization basins and combined sewer storage) to moderate extreme loads, potentially stabilizing emissions.[36]

Influence of Sludge Factors. Sludge yield was positively correlated with predicted emissions (Figure 3e), as higher sludge production (>1 × 10^4 tons) generally reflected greater organic and nutrient loads that increase energy demand and indirect emissions. Sludge yield ranked fourth regarding scope 2 emissions. Potential reduction should be attempted to improve aeration efficiency, advancing sludge reduction and anaerobic digestion, thereby significantly reducing indirect emissions.[37]

Influence of Process Configuration. The ranking of pollutant removal (COD_{rem} accounted for 6.65% of CO_2 and 7.16% of CH_4) highlighted the influence of refined biochemical process control and process type.[38] Although process configuration was affected by collinearity and therefore was not included as a model input, it still had potential relevance to GHG intensity. Multivariate clustering (Figure 4b,c, with the selected key predictors and predicted emissions) could identify categories of WWTPs with similar influent quality, treatment scale, and operational characteristics (based on 2019 data),[39] showing cluster 3 (high-pollution, high-emission type, mean = 1.08 kg CO_2-e/m^3) > cluster 0 (0.50) > cluster 1 (0.43) > cluster 2 (0.39) (Table S37).

A^2O was widely distributed across clusters (Table S38), while OD (27.4%) and A^2O (33.5%) had relatively large proportions and SBR was comparatively low (11.8%) in cluster 2. In cluster 3, the proportion of SBR increased to 20.8%, suggesting that SBR was more commonly used under high-TN conditions.[9,40] Boxplot analysis (Figure 4d) showed that SBR had the highest GHG emissions (0.73 ± 0.39 kg CO_2-e/m^3), possibly due to transient N_2O release and aeration start-up losses induced by cyclic operation. In addition, CAS (0.48 ± 0.36 kg CO_2-e/m^3) and A^2O (0.47 ± 0.30 kg CO_2-e/m^3) exhibited comparable emission levels, whereas OD showed the lowest emissions (0.43 ± 0.29 kg CO_2-e/m^3) owing to its more stable aerobic conditions and higher hydraulic retention time.

Spatial Heterogeneity Drivers. Latitude acted as a mediating feature (Figure 3f), and its high ranking further confirmed the necessity of the *PROV_type_model*. Collinearity analysis (VIF = 3.97 < 5) and partial correlation analysis (+0.216 reduced to +0.041) indicated that latitude contributed to emissions both directly and indirectly. SEM incorporating the key WWTP predictors was employed to capture the main structural relationships among variables (Figure 4a, comparative fit index, CFI = 0.984; goodness-of-fit index, GFI = 0.993; root mean square error of approximation, RMSEA = 0.075). The total effect of latitude reached r = 0.216 (direct effect r = 0.111, indirect effects via COD_{inf} and TN_{inf} of 0.056 and 0.049, all p < 0.001), which highlighted that regional variation influenced predicted emissions through both direct geographic and indirect process-mediated pathways (impacting COD_{inf} and TN_{inf}).[3] Although latitude was not a controllable factor, subtropical regions (25−30°N) exhibited relatively lower predicted emissions.

GHGs' Spatial Heterogeneity Index Reflecting Urban Development Patterns

Figure 4e,f showed the linear regression analysis among GDP, population, and EF-derived cumulative GHG emissions (R^2 = 0.518/0.439), demonstrating that WWTPs have evolved alongside urban economic development and population growth as a critical component linking natural water bodies and domestic water use.[41] We introduce the GHG_SHI, a novel indicator that linked EF-derived GHG inventories and urban development.[16] Overall, N_2O and EC exhibited the strongest interprovincial heterogeneity (median SHI = 0.99 and 0.86), whereas CO_2 and CC were the most homogeneous (median SHI = 0.24 and 0.16). Specifically, provinces such as Ningxia (GHG_SHI = 0.06) and Shanghai (0.18) exhibited the lowest values for N_2O, suggesting highly consistent and predictable emission profiles—indicative of standardized operations and coherent management frameworks (Figure 5a). In contrast, provinces including Fujian (2.62), Jiangxi (2.20), and Guangdong (1.72) demonstrated markedly elevated GHG_SHI values, revealing pronounced spatial heterogeneity and irregular emission behaviors, likely resulting from diversified technological deployments and varying operational practices. It was noteworthy that Guangdong maintained a high SHI even with the largest sample size (n = 4443, Figure S37)—its bootstrap confidence interval was relatively narrow (such as scope 1 emission), demonstrating that the heterogeneity was not a result of estimation noise but rather reflected genuine interplant differences within wastewater treatment facilities of a megaregion with 125 million people and a GDP of 10.8 trillion CNY.

As for climate and environmental conditions (Figures 5b−g), the N_2O SHI exhibited significant correlations with multiple climatic factors. Sunshine duration (SD) showed the strongest negative correlation with N_2O SHI (PCC = −0.42, R^2 = 0.18, p < 0.05): provinces with longer SD displayed lower N_2O heterogeneity—for example, northern provinces such as Jilin (SD = 3010.6 h, SHI = 0.45) and Tibet (2955.1 h, 0.51) all had lower N_2O SHI values, whereas southwestern provinces with shorter SDs exhibited higher heterogeneity [such as Chongqing (964.5 h, 2.07); Sichuan (858.5 h 1.06). Rainfall (RFA, PCC = +0.41, R^2 = 0.17) and relative humidity (HD, PCC = +0.41, R^2 = 0.17) were positively correlated with N_2O SHI: the northwestern provinces with low humidity and precipitation exhibited the lowest N_2O heterogeneity (Ningxia HD = 48%, RFA = 145.5 mm, N_2O SHI = 0.06; Tibet HD = 37%, N_2O SHI = 0.51). It was suggested that N_2O GHG_SHI could serve as a feedback indicator for solar resource levels: provinces with longer sunshine generally exhibited lower spatial heterogeneity, reflecting how solar exposure influenced variability in N_2O emissions across WWTPs through differential adoption and efficiency of solar-driven processes (e.g., photodegradation, solar-aeration).[42]

Among water resource utilization and management indicators, per capita water usage (PCWU) exhibited consistent positive correlations with CC GHG_SHI (PCC = 0.08−0.45; R^2 = 0.20, p < 0.05). Provinces with higher individual water demand, such as Xinjiang (2346.1 m^3/person) and Ningxia (1010.8 m^3/person), exhibited significantly higher CC SHI (Xinjiang 0.57, Tibet 0.49). Similarly, water resources (WRs) also showed positive associations with CC GHG_SHI (PCC = 0.45, R^2 = 0.20): the most water-rich regions, Tibet (465.82 billion m^3/year) and Sichuan (295.26 billion m^3/year), had CC SHI values of 0.49 and 0.69, respectively,

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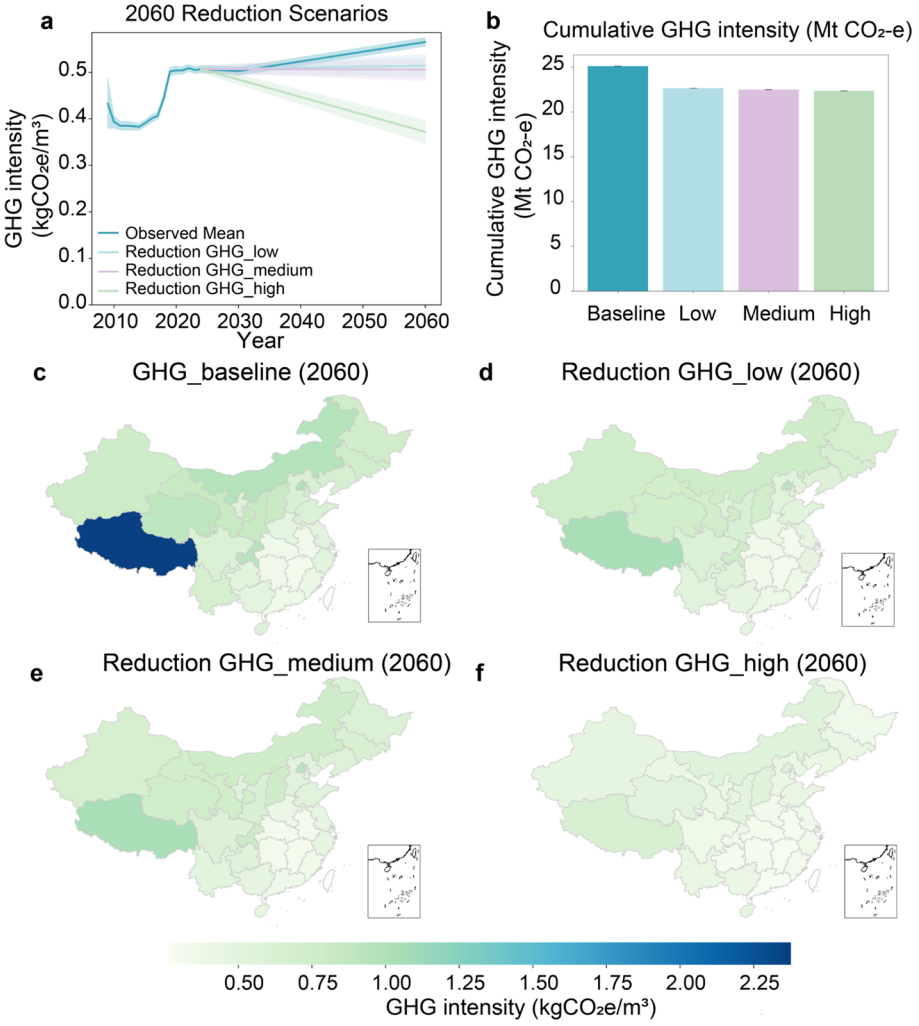


Figure 6. WaterMAP-assisted exploratory sensitivity analysis toward 2060: (a,b) future trends in GHG intensity under baselines and WaterMAP-guidance and (c–f) GHG intensity under baselines and WaterMAP guidance in 2060.

reflecting that areas with abundant but unevenly distributed water resources often experience more irregular chemical dosing and treatment practices.[43] Total domestic water consumption (TDWC) showed the strongest association with EC (PCC = +0.41, R^2 = 0.16, p < 0.05): provinces with the largest water use, such as Jiangsu (61.91 billion m^3), Xinjiang (58.77), and Guangdong (41.23), generally exhibited higher EC SHI (Jiangsu 1.27, Xinjiang 1.25, Guangdong 2.02), suggesting that provinces with larger water-use scales display greater interplant differences in EC emissions.

Regarding socioeconomic development and infrastructure investment indicators, water facility construction land area (WFCLA) was positively correlated with GHG_SHI except for CO_2 (0.02–0.50), indicating that provinces with smaller water infrastructure footprints (such as Beijing 61.7 ha, the lowest WFCLA with all the GHG_SHI <0.52) tended to have higher spatial homogeneity in carbon emissions. Moreover, elevated GHG_SHI in the regions might reflect the complexity and unevenness of the distributed water treatment and reuse systems. Population and GDP were both significantly positively correlated with EC-related SHI (PL: PCC = +0.40, R^2 = 0.16; GDP: PCC = +0.35, R^2 = 0.12): provinces with the largest populations, such as Guangdong (125 million), and the highest GDP, Guangdong (10.8 trillion CNY) and Jiangsu (9.87 trillion CNY), exhibited EC SHI values of 2.02 and 1.27, respectively. This suggested that provinces with larger populations and economic scales display stronger interplant differences in EC emissions, possibly reflecting the combined effects of varying urban development levels and power grid structures within large provinces. Anomalous SHI values observed among provinces with similar socio-economic levels might reflect internal heterogeneity in infrastructure distribution or operational management, highlighting the diagnostic and feedback value of SHI in regional governance.[44,45]

Among environmental sustainability and risk management parameters, sudden environmental incidents (SEIs) exhibited the strongest positive correlation with CO_2 SHI (PCC = +0.40, R^2 = 0.16, p < 0.05): provinces without any recorded environmental incidents throughout the year, such as Tibet, Hainan, and Yunnan, displayed low CO_2 SHI (0.06–0.16). DWCV was significantly positively correlated with CH_4 and EC SHI (CH_4: PCC = +0.40, R^2 = 0.16; EC: PCC = +0.46, R^2 = 0.21, p < 0.05): provinces with larger waste streams, including Guangdong (33.47 million tons), Jiangsu (18.09), and Shandong (17.87), exhibited high EC SHI values (Guangdong 2.02, Jiangsu 1.27, Shandong 1.37). The domestic waste nonhazardous treatment rate (DWNH) showed weak negative correlations with N_2O, CH_4, and EC SHI (PCC =

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−0.15 to −0.20): Chongqing (88.8%) and Jilin (90.2%) displayed a higher heterogeneity (Chongqing N_2O SHI = 2.07), reflecting that uneven environmental management systems may amplify spatial differences in emissions. From a diagnostic perspective, these elevated SHI values indicate internal fluctuations in wastewater treatment operations and environmental governance systems.[46]

As a result, the spatial patterns of GHG emissions from WWTPs (provincial homogeneity/heterogeneity) were not merely products of environmental processes but also served as diagnostic indicators of underlying natural and socioeconomic systems. Incorporating WWTPs into urban material-energy metabolism networks could not only enhance energy self-sufficiency but also provide a crucial dynamic basis for assessing regional low-carbon transition strategies.[47]

Scenario Analysis

WaterMAP-Assisted Exploratory Sensitivity Analysis toward 2060. We further predicted the volumetric GHG intensity for 2060 (0.565 kg CO_2-e/m^3) using *NAT_total_model* (Figure 6a,b), indicating greater long-term challenges. Based on the key environmental predictor ranges, three scenario pathways with different intensities (low, medium, high) were constructed and compared with the baseline. Low scenario applied mild optimization (COD_{inf} 200−250 mg/L, TN_{inf} 15−25 mg/L, ∼5% sludge reduction, ∼5% volume increase) and medium applied stricter source control (COD_{inf} 100−200 mg/L, TN_{inf} 0−20 mg/L, ∼25% sludge reduction, medium volume). High scenario represented strong model-guided optimization and intervention, simulating idealized high-level pollutant control and operational modes. In the key predictors of EF-derived emissions, COD_{inf} and TN_{inf} were primarily controlled by upstream watershed processes and hard to regulate by WWTPs. The sensitivity analysis could provide insights for large-scale policy design and emission standard optimization. Sludge yield and treatment capacity were partially controllable variables, whose regulation required a trade-off among cost, process constraints, and effluent quality requirements. The volumetric GHG intensity under each scenario decreased to 0.514 [0.509, 0.520], 0.506 [0.502, 0.510], and 0.371 [0.370, 0.375] kg CO_2-e/m^3, respectively (Figure 6c−f), with EC and N_2O remaining dominant contributors (0.295 and 0.050 kg CO_2-e/m^3) via *NAT_type_model* prediction. Even under mild control, the sector could still achieve approximately 9.6%−10.4% GHG reduction. In contrast, the optimal high-intensity scenario could reduce GHG intensity by around 34%, while total emissions would also decrease by about 10.8%, reaching 22.4 Mt CO_2-e. This highlighted the substantial potential of implementing stringent source control and deep optimization of operational modes within the defined system boundaries.

Transfer Learning Enhanced GHG Mitigation Policies. The transfer-performance matrix showed clear asymmetry and revealed characteristic high-transferability clusters (Figure S41 and Text S7) for N_2O (Shanghai, Shanxi, Ningxia, and Shandong) and EC (Tianjin, Jiangsu, Liaoning, and Inner Mongolia). The high-transferability patterns indicated that China's WWTPs have formed three technological ecosystems—standardized coastal regions, engineering-stable northern industrial belts, and adaptive systems in western ecologically fragile areas. Moreover, high-transferability *PROV_type_model* demonstrated much lower model uncertainty (e.g., EC model certainty: 91−94%) compared with *NAT_type_model* (56%), highlighting the superior generalizability (Figures S42−S47 and Tables S48, S49). The scenario suggested practical pathways for cross-province GHG mitigation, including collaborative low-carbon technology deployment within high-transfer clusters and the development of interprovincial technical-support and carbon-market mechanisms to advance coordinated decarbonization across China's WWTPs.

■ ENVIRONMENTAL IMPLICATIONS

In this study, we developed WaterMAP, a plant-level modeling framework combined with EF inventories following a scalable multi-GHG paradigm for wastewater treatment. Overall, *NAT_total_model* achieved robust national-scale prediction (R^2 = 0.60; RMSE = 0.19 kg CO_2-e/m^3), while *NAT_type_model* (R^2 ranging from 0.54 (N_2O) to 0.99 (CC)) and *PROV_type_model* progressively improved GHG-specific and regional prediction, supporting a scalable modeling strategy to address national accounting, GHG-specific attribution, and regional analysis. The rigorous validation strategy was used to assess model generalization across unseen time periods, provinces, WWTPs, and independent data sets (Text S6). LOYO tests showed that EF-derived CC, CO_2, and EC retained near-original temporal performance (R^2 relative decline <5%), and CH_4 exhibited moderate temporal generalization similar to total emissions (17.3% vs 17.2%), while only N_2O dropped markedly. Out-of-sample evaluation also using EF-derived emissions revealed that *NAT_Type_Model* predicted total emissions with a MAPE of 12.6% (Figure S39), with errors lower than most individual GHG components, indicating WaterMAP's transferability and reflecting partial error cancellation in aggregated emissions. While these results demonstrate WaterMAP's extrapolation capability, LOPO tests confirmed limited province-level generalization for *NAT_Total_Model* and *NAT_Type_Model*, justifying the development of *PROV_Type_Model* and the use of transfer learning to capture spatial heterogeneity.

Compared with previous GHG estimation studies[2,3,6,9,22,40,48] (Table S39), which mainly reported national-scale baseline inventories (e.g., direct emissions in the US of ∼47.0 Mt CO_2-e/year[2]), this study combined EF-derived accounting and ML by incorporating multifactor predictive capability (RMSE = 0.19 kg CO_2-e/m^3) and uncertainty assessment, thereby capturing nonlinear and high-dimensional interactions affecting EF-derived inventories. Compared with ML-based studies on GHG prediction[15,49−58] (Table S40), which mainly focused on forecasting EC (with R^2 up to 0.68)/N_2O emissions (R^2 up to 0.94) or optimizing chemical dosing, previous models were typically applied to individual WWTPs/specific process units/single GHG source. WaterMAP integrated tree-based models and advanced DL foundation models to achieve large-scale scalable and comprehensive accounting of full-scope GHG emissions. Unlike single-process models, it covered the entire emission boundary of WWTPs; unlike macro-scale national inventories, it could predict future GHG intensity with high accuracy across scales and identify key multiscale predictors.

In practical environmental applications, *NAT_total_model* and *NAT_type_model* identified key predictors of EF-derived emissions, particularly the hydraulic load (volume), water quality parameters (TN_{inf}, COD_{inf}), latitude, and process. Although the factors are hard to control in real time, they can provide guidance for designing and selecting reduction

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pathways for new or retrofitted WWTPs. Real-time WaterMAP control could further integrate sensors, automated process control, and large language models (LLMs) in the future, incorporating additional actionable environmental factors (e.g., dissolved oxygen). *PROV_type_model*, in addition to providing finer-grained predictions, enabled diagnostic evaluation of urban development indicators using GHG_SHI, while transfer learning further guided future cross-regional coordinated GHG management.

This study also has several limitations and future prospects: (1) the current framework has been developed and validated within the Chinese WWTP context using EF-derived labels. Application to other countries, treatment configurations, or measured emission regimes would require region-specific calibration and validation; (2) EF-derived inventories may inherit assumptions, regional biases, and process-specific uncertainties from accounting protocols, which can also be reproduced by ML models.[59] Therefore, site-specific GHG measurements are needed to calibrate and validate EF-based estimates, while future work should integrate localized EFs with onsite measurements to improve accounting accuracy; and (3) future work could couple WaterMAP-identified driving factors with process-based optimization frameworks (e.g., ASM-coupled cost-benefit optimization, multiobjective optimization) and integrate LLM techniques with emerging wastewater treatment and resource recovery technologies to support the transition of WWTPs from treatment units to resource-recovery facilities, fundamentally reshaping system-level carbon-metabolism pathways.[60]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.6c03422.

Additional contents about data collection, data preprocessing, WaterMAP model development, model evaluation, hyperparameters tuning, process construction, EF-based inventories, model interpretation (SHAP, PDP, SEM, and VIF), uncertainty analysis, generalization analysis, and transfer learning (DOCX)

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Notes

The authors declare no competing financial interest.

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